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Phylum (class)	Examples	Tissue organization	Coelom	Embryonic development	Body symmetry	Important facts
Porifera	Sponge 	No true tissues	-	-	Asymmetrical	Sessile (non-motile), aquatic habitats, most primitive animals
Cnidaria	Jellyfish, hydra, coral 	Diploblasts	-	-	Radial	Aquatic habitats
Platyhelminthes	Flatworm, tapeworm 	Triploblasts	Acoelomate	Protostome	Bilateral	Parasitic, most primitive of triploblasts, have organs, non-segmented worms
Nematoda	Roundworm, hookworm 	Triploblasts	Pseudo-coelomate	Protostome	Bilateral	Parasitic, non-segmented worms
Rotifera	Rotifer 	Triploblasts	Pseudo-coelomate	Protostome	Bilateral	Non-segmented, aquatic zooplankton
Annelida	Earthworm, leech 	Triploblasts	Coelomate	Protostome	Bilateral	First segmented phyla, closed circulatory system
Mollusca	Clam, snail, squid, octopus 	Triploblasts	Coelomate	Protostome	Bilateral	Mostly aquatic, soft-bodied, can have shells
Arthropoda (Insecta, Arachnida, Crustacea)	Ant, spider, lobster 	Triploblasts	Coelomate	Protostome	Bilateral	Most diverse phylum, highest number of species, chitin exoskeleton, jointed appendages, segmented bodies, open circulatory system (w/ hemolymph)
Echinodermata	Starfish, sea urchin 	Triploblasts	Coelomate	Deuterostome	Radial (adult)	Closest related major phyla to chordates
Chordata	Vertebrates 	Triploblasts	Coelomate	Deuterostome	Bilateral	Segmented bodies, notochord, dorsal hollow nerve cord, pharyngeal pouches (slits), post-anal tail

DAT Bootcamp High-Yield Chordata Classification | Bootcamp.com

Subphylum	Class	Examples	Important facts
Invertebrates	N/A	Tunicate, lancelet 	Same developmental characteristics as other chordates, lack vertebrae
Vertebrata	Jawless fish (agnatha)	Lamprey, hagfish 	Vertebrae, cartilaginous skeleton, breathe via countercurrent exchange through gills
Vertebrata	Cartilaginous fish (chondrichthyes)	Shark 	Vertebrae, cartilaginous skeleton, gills, jaws and teeth
Vertebrata	Bony fish (osteichthyes)	Salmon, halibut 	Vertebrae, jaws, gills, bony skeleton; lobe-finned fish subcategory developed muscular fins resembling limbs
Vertebrata	Amphibia	Tadpole, frog, toad, newt, salamander 	Vertebrae, jaws, lungs, limbs with digits
Vertebrata	Reptilia	Turtle, snake 	Vertebrae, jaws, lungs, limbs, amniotic eggs *Crocodilians have four-chambered hearts
Vertebrata	Birds	Eagle, blue jay 	Vertebrae, jaws, lungs, limbs, amniotic eggs, four-chambered hearts
Vertebrata	Mammalia	Kangaroo, whale, human 	Vertebrae, jaws, lungs, limbs, amniotic eggs, four-chambered hearts, mammary glands, diaphragms, hair

DAT Bootcamp High-Yield Plant Classification | Bootcamp.com

Clade name	Key phyla	Examples	General traits	Keywords
Nonvascular plants (bryophytes)	Bryophyta	Hornworts, mosses, liverworts 	- No vascular tissue - Gametophyte dominant life cycles - Reduced sporophyte is dependent on gametophyte for survival - Flagellated sperm - Rhizoids anchor gametophytes - Very little vertical growth - Exist in moist habitats	Rhizoids anchor gametophytes (no roots)
Seedless vascular plants (tracheophytes)	Lycophyta, Pterophyta	Club moss, quillworts, fern, horsetail 	- Contain vascular tissue (xylem and phloem) - Flagellated sperm - Exist in moist habitats - Most are homosporous - Independent gametophyte & sporophyte life cycles	- Roots: Structural support and water absorption - Leaves: Improved photosynthesis
Seeded vascular plants (tracheophytes)	Gymnosperms	Fir, spruce, redwood, pine, conifers 	- Contain vascular tissue - Unprotected seeds - Non-flagellated sperm - Pollen dispersed by wind - Heterosporous - Water and nutrient transport makes survival in more arid environments possible	- Pollen: Improved fertilization efficiency - Seeds: Improved dispersal and durability
	Angiosperms	Flowering plants, magnoliophyta 	- Contain vascular tissue - Seeds protected in fruit (ovaries) - Non-flagellated sperm - Heterosporous - Most abundant type of plant - Exhibit double fertilization - Water and nutrient transport makes survival in more arid environments possible	- Flowers and fruits - Double fertilization